



using resources

Name: _____

Class: _____

Date: _____

Time:

Marks: **66 marks**

Comments:

1.

This question is about drinking water.

There are two main steps in producing drinking water from fresh water.

(a) Draw **one** line from each step to the reason for the step.

Step	Reason for step
	Desalination
Filtration	Improve taste
	Increase pH
Sterilisation	Kill bacteria
	Remove solids

(2)

(b) Which **two** substances are used to sterilise fresh water?

Tick (✓) **two** boxes.

Ammonia

☐

Chlorine

☐

Hydrogen

☐

Nitrogen

☐

Ozone

☐

(2)

A large amount of aluminium sulfate was accidentally added to the drinking water supply at a water treatment works.

- (c) Scientists tested a sample of the drinking water to show that it contained dissolved solids.

Which **two** methods show the presence of dissolved solids in the sample of drinking water?

Tick (✓) **two** boxes.

Add damp litmus paper to the sample.

☐

Evaporate all water from the sample.

☐

Measure the sample's boiling point.

☐

Test the sample with a glowing splint.

☐

(2)

- (d) Scientists tested two water samples from the drinking water supply.

The scientists tested one sample for aluminium ions and the other sample for sulfate ions.

Draw **one** line from each ion to the compound needed to identify the ion.

Ion	Compound needed to identify ion
	Barium chloride
Aluminium ion	Copper sulfate
	Silver nitrate
Sulfate ion	Sodium hydroxide
	Sulfuric acid

(2)

- (e) How could pure water be produced from drinking water that contained dissolved solids?

Tick (✓) **one** boxes.

Chromatography

☐

Cracking

☐

Distillation

☐

Sedimentation

☐

(1)

(Total 9 marks)

2.

A large amount of aluminium sulfate was accidentally added to the drinking water supply at a water treatment works.

- (a) Describe a test to show that the drinking water contained aluminium ions.

Give the result of the test.

Test _____

Result _____

(3)

- (b) Describe a test to show that the drinking water contained sulfate ions.

Give the result of the test.

Test _____

Result _____

(2)

- Your investigation should produce valid results.

[illegible]

(Total 9 marks)

Disposable cups are made from coated paper or poly(styrene).

	Coated paper cups	Poly(styrene) cups
Raw materials	Wood	Crude oil
Mass of 1 cup in g	8.3	1.9
Energy to produce 1 cup in kJ	550	200
Energy released when 1 cup is burned in kJ	166	76
Biodegradable	Yes	No
Recyclable	No	Yes

- (a) Evaluate the use of coated paper compared with poly(styrene) to make disposable cups.

Use the table above and your knowledge and understanding of LCAs.

(6)

- (b) Calculate the energy needed to produce 1.00 kg of coated paper cups.

Use the table above.

Give your answer in standard form.

Energy = _____ kJ

(2)

- (c) Melamine is a polymer used to make non-disposable cups.

Melamine does **not** melt when it is heated.

Explain why.

(2)

(Total 10 marks)

4.

Fertilisers are used to improve agricultural productivity.

- (a) Ammonium nitrate is used in fertilisers.

Name the **two** compounds used to manufacture ammonium nitrate.

(1)

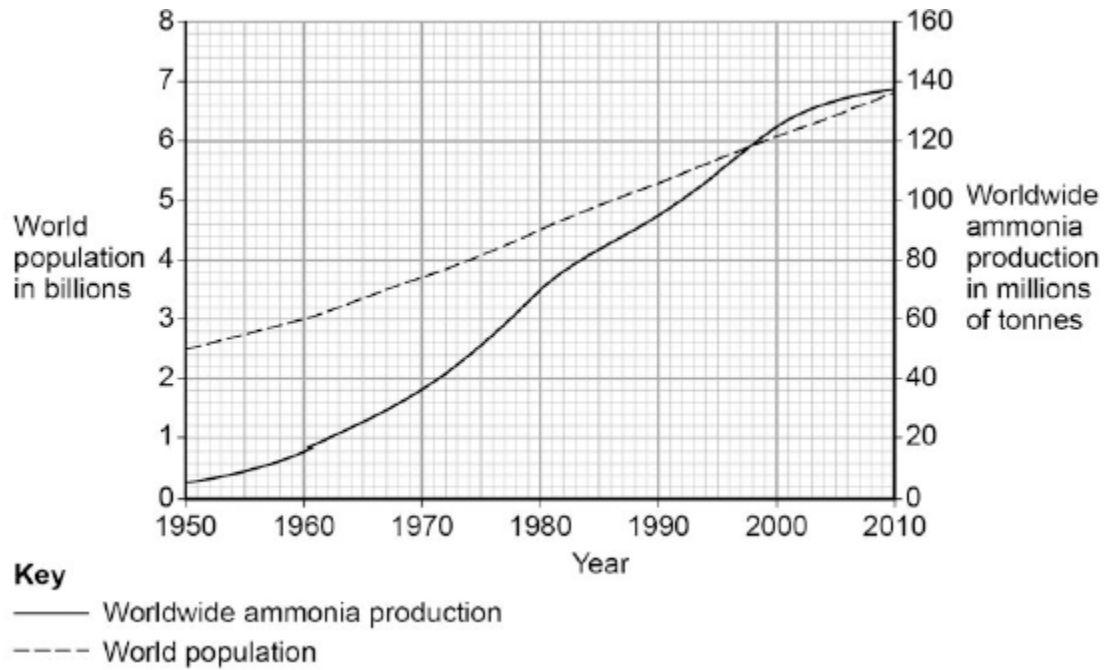
- (b) A fertiliser contains the following information on the label:

NPK value = 14 : 11 : 11

Explain why this information is useful to farmers.

(2)

- (c) The figure below shows worldwide ammonia production and world population from 1950 to 2010.



Use the figure above and your knowledge to explain the relationship between ammonia production and world population.

(3)
(Total 6 marks)

5.

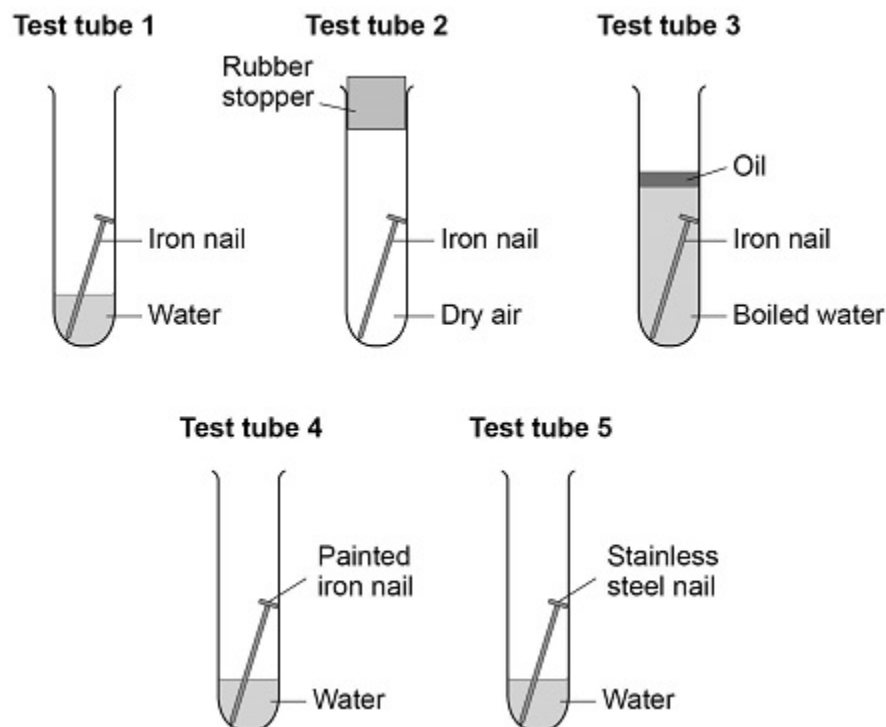
This question is about the corrosion of metals.

The corrosion of iron is called rusting.

(a) A student investigated the rusting of iron.

This is the method used.

1. Set up the test tubes as shown in the figure below.
2. Leave the test tubes for 1 week.
3. Examine the nails for signs of rust.



Explain what would happen to the nails in each of the test tubes.

(5)

- (b) Magnesium is fixed to some steel ships.

Explain how this prevents the steel from rusting.

(2)

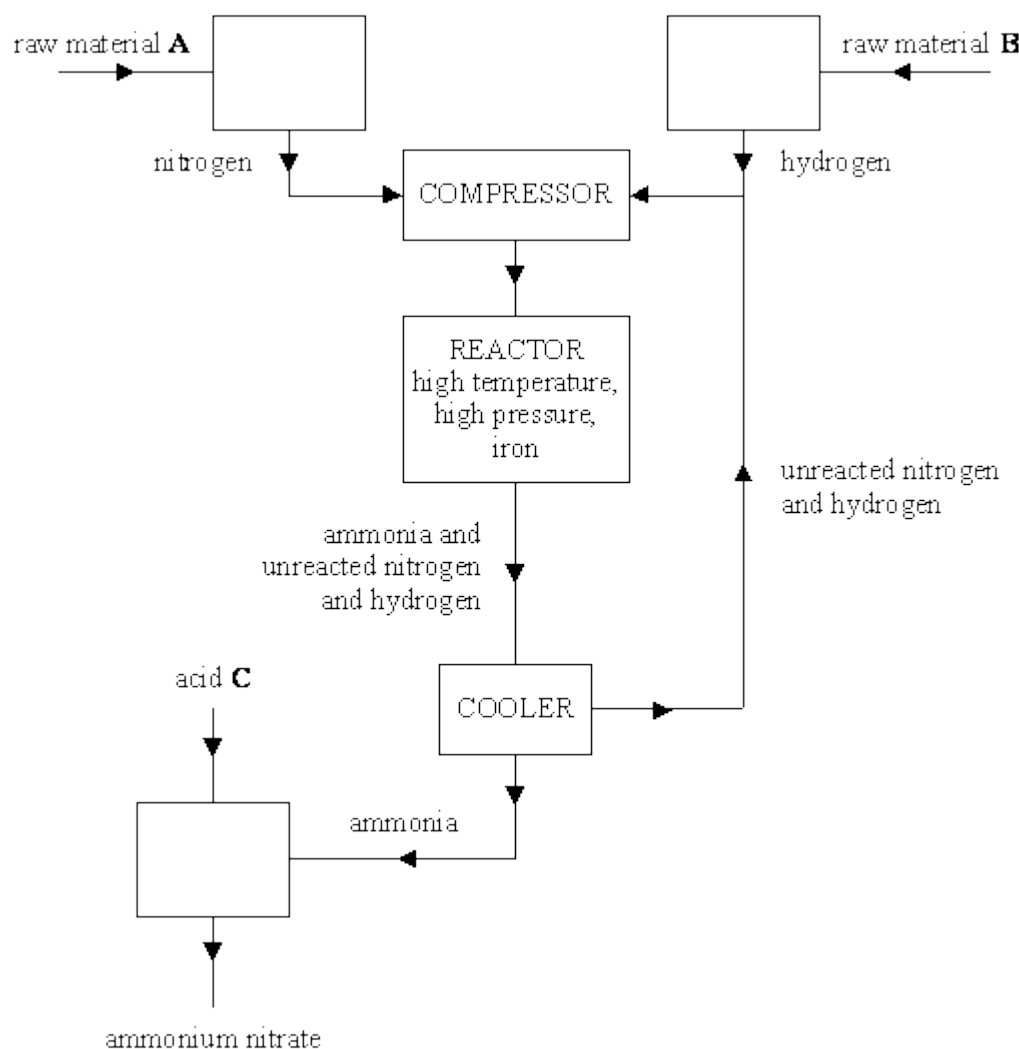
- (c) Explain why aluminium window frames do **not** corrode after they are made.

(2)

(Total 9 marks)

6.

The flow chart below shows the main stages in the production of ammonium nitrate.



- (a) (i) Name the two raw materials shown in the flow chart as **A** and **B**.

Raw material **A** _____

Raw material **B** _____

(2)

- (ii) What is the purpose of the iron in the reactor?

(1)

- (b) (i) Balance the equation which represents the reaction which produces ammonia in the Haber process.



(1)

- (ii) The table shows how temperature and pressure affect the amount of ammonia produced in this reaction.

TEMPERATURE (°C)	PRESSURE (ATM)	PERCENTAGE OF NITROGEN AND HYDROGEN CONVERTED TO AMMONIA (%)
250	200	75
250	1000	96
1000	1	0.01
1000	1000	1

Explain, as fully as you can, why a temperature of about 450°C and a pressure of about 200 atmospheres are normally used in the industrial process.

(6)
(Total 10 marks)

7.

Figure 1 shows a surfer on a surfboard.

Figure 1



Surfboards are made from polymers.

Surfboards have a poly(styrene) core and an outer skin.

(a) **Figure 2** shows the displayed structural formula of poly(styrene).

Figure 2

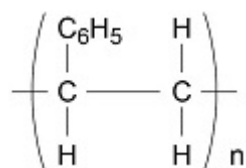
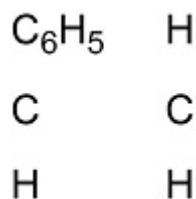


Figure 3 shows an incomplete displayed structural formula of the monomer styrene.

Complete **Figure 3**.

Figure 3



(2)

The outer skin of surfboards contains a polyester.

Two monomers, **A** and **B**, are needed to make the polyester.

Figure 4 shows how these two monomers are represented.

Figure 4



- (b) Name the functional group in monomer **B**.

(1)

- (c) Monomers **A** and **B** join together to produce a polyester and a small molecule.

Name the small molecule.

(1)

- (d) Why does this type of polyester melt when it is heated?

(2)

The outer skin of surfboards is a composite material.

The composite material contains glass fibres surrounded by a polyester.

(e) Draw **one** line from each material to the description of that material.

Material	Description of the material
	Hydrocarbon
Glass fibres	Matrix
	Monomer
Polyester	Polypeptide
	Reinforcement

(2)

(f) The outer skin makes the surfboard more expensive.

Suggest **two** reasons why an outer skin is added to the poly(styrene) core.

1 _____

2 _____

(2)

(Total 10 marks)

8.

Read the article and then answer the questions.

Supermarkets launch eco-friendly plastic milk bags. Could this be the end of the milk bottle?



Milk bottles are made from glass or from plastic.

Glass milk bottles contain 0.5 litres of milk. When the milk is used up the empty bottles are returned to be re-used. Glass milk bottles are re-used 24 times on average. The glass to make new milk bottles is produced when a mixture of sand, limestone, soda and recycled glass is heated to about 1600 °C in a furnace. There are almost unlimited amounts of the raw materials needed to produce this glass. About 35% of used glass is recycled.

The most common plastic milk bottles contain 2 litres of milk. When the milk is used up the empty bottles are discarded as waste. The plastic used to make these milk bottles is poly(ethene). Poly(ethene) is produced from crude oil by first using fractional distillation, then cracking the naphtha fraction and finally polymerising the ethene. About 5% of used poly(ethene) is recycled.

The new plastic milk bags contain 2 litres of milk. The milk bags are also made from poly(ethene). A milk bag uses 75% less poly(ethene) than is used to make the poly(ethene) milk bottles. When the milk is used up the empty bags are discarded as waste.

- (a) Describe what happens in fractional distillation so that fractions, such as naphtha, are separated from crude oil.

(3)

- (b) Supermarkets claim that using milk bags instead of milk bottles would have less environmental impact.

Do you agree with this claim?

Use the information in the article and your knowledge and understanding to make appropriate comparisons to justify your answer.

This image shows a blank sheet of white paper with horizontal ruling lines. The lines are evenly spaced and extend across the width of the page. There are no margins, text, or other markings on the paper.

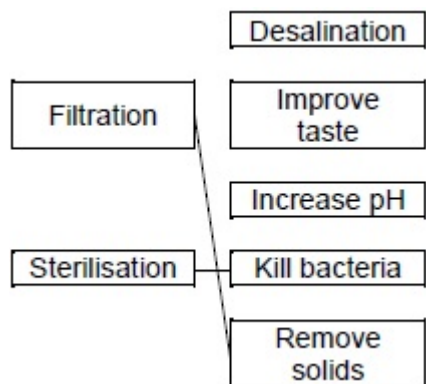
(4)

(Total 7 marks)

Mark schemes

1.

(a)



*an extra line from a step to a reason for that step
negates that mark*

2

(b) chlorine

1

ozone

1

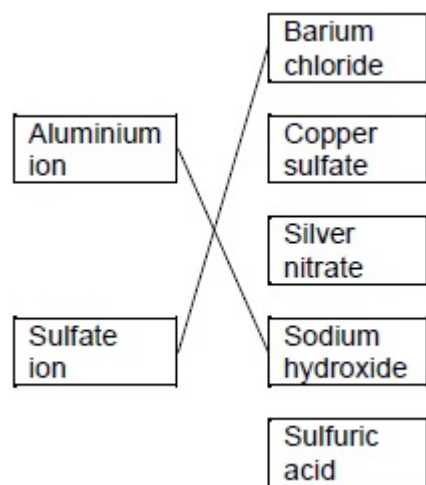
(c) evaporate all water from the sample

1

measure the sample's boiling point

1

(d)



*an extra line from an ion to a compound needed
negates that mark*

2

(e) distillation

1

[9]

2.	(a) add sodium hydroxide (solution to water sample)	1
	white precipitate (forms) <i>dependent on correct test in MP1</i>	1
	(precipitate which is) soluble in excess (NaOH) <i>dependent on correct test in MP1</i>	1
	(b) add barium chloride (solution) and (dilute) hydrochloric acid (to water sample) <i>allow barium nitrate (solution)</i> <i>allow (dilute) nitric acid</i>	1
	white precipitate (forms) <i>dependent on addition of barium chloride / nitrate (solution) in MP1</i>	1
	(c) Level 2: The design/plan would lead to the production of a valid outcome. All key steps are identified and logically sequenced.	3–4
	Level 1: The design/plan would not lead to a valid outcome. Some relevant steps are identified, but links are not made clear.	1–2
	No relevant content	0
	Indicative content	
	- weigh (evaporating) basin / dish - add measured volume of water - weigh (evaporating) basin / dish and water - heat to evaporate water - reweigh - repeat heating until constant mass obtained - subtract mass of (evaporating) basin / dish from mass - repeat and calculate a mean, discarding anomalous results - calculate the mass in 100 cm³ water if necessary	[9]

3.

- (a) **Level 3:** A judgement, strongly linked and logically supported by a sufficient range of correct reasons, is given.

5–6

Level 2: Some logically linked reasons are given. There may also be a simple judgement.

3–4

Level 1: Relevant points are made. They are not logically linked.

1–2

No relevant content

0

Indicative content

raw materials

- crude oil finite **or** will run out (so will be unavailable for other uses)
- wood is a renewable resource
- wood involves land use for forestry (so less available for agriculture / food)
- wood may involve deforestation (so reduces biodiversity)

manufacturing

- both require energy which may be derived from finite fuels (so they run out more quickly)
- paper more energy intensive (so more pollution is possible)
- the need for more energy for paper potentially releases more carbon dioxide to the atmosphere (so increases global warming)
- paper involves higher water usage (so increases the potential for water pollution)
- paper cups are heavier to transport (so have higher energy requirement)
- packaging requirements similar (so neither has an advantage)

usage

- both single-use (so neither has an advantage)

disposal

- paper releases more energy if incinerated (so more energy can be used for other purposes)
- paper will decompose (so will not remain in landfill)
- poly(styrene) could release toxins on incineration
- poly(styrene) will not decompose (so will remain in landfill)
- poly(styrene) can be used to manufacture other products (so conserves energy **or** finite resources)
- both can cause litter **or** visual pollution

(b) $\frac{1000}{8.3} \times 550 \text{ (kJ)}$

1

$= 6.63 \times 10^4 \text{ (kJ)}$

allow $6.6265060240963 \times 10^4 \text{ (kJ)}$ correctly rounded

allow 66265.060240963 (kJ) correctly rounded for 1 mark

1

an answer of $6.63 \times 10^4 \text{ (kJ)}$ scores 2 marks

(c) (melamine is a) thermosetting (polymer)

1

(which) contains crosslinks / bonds (between polymer chains)

*do **not** accept reference to intermolecular forces*

allow (so) it decomposes

1

[10]

4.

(a) ammonia **and** nitric acid

allow NH_4OH

allow $\text{NH}_3(\text{aq})$

1

(b) shows fertilisers are formulations

allow gives percentage / proportion of nitrogen, phosphorus and potassium in the fertiliser

1

(so) farmers can choose fertiliser with required properties

1

(c) as world population increases, ammonia production increases

1

ammonia is used to produce fertilisers

1

so increasing need for fertilisers as more food required for increased population

allow as more food produced less mortality

1

[6]

5.

- (a) Tube 1: (nail) rusts because air / oxygen **and** water present

1

Tube 2: (nail) does not rust because no water

allow Tube 2: (nail) does not rust because only air / oxygen

1

Tube 3: (nail) does not rust because no air / oxygen

allow Tube 3: (nail) does not rust because only water

1

Tube 4: (nail) does not rust because paint is a barrier (to water / air / oxygen)

allow Tube 4: (nail) does not rust because paint is a protective layer / coating (against water / air / oxygen)

or

allow Tube 4: (nail) does not rust because paint protects it from water / air / oxygen

1

Tube 5: (nail) does not rust because stainless steel resistant to corrosion

allow Tube 5: (nail) does not rust because stainless steel does not corrode

allow Tube 5: (nail) does not rust because stainless steel contains nickel / chromium

1

If no other mark awarded allow 1 mark for correct rusting pattern in all 5 tubes

- (b)

allow converse

magnesium is more reactive (than iron)

allow magnesium is more reactive (than steel)

1

(so magnesium) provides sacrificial protection

allow (so magnesium) corrodes / reacts instead of iron / steel

allow (so magnesium) corrodes / reacts before iron / steel

ignore references to protective layers

ignore references to magnesium rusting

1

- (c) (aluminium has a coating of) aluminium oxide

1

(so the aluminium oxide) protects the metal (from further corrosion)

allow (so aluminium oxide) prevents water / air / oxygen from reaching the metal

1

[9]

6.

- (a) (i) A = air
B = natural gas / methane / north sea gas / CH₄ / oil /
naphtha/ steam water (H₂O)

Accept answers written in the box at the start of the question.
each for 1 mark

2

- (ii) catalyst / speed up the reaction / lower the activation energy
for 1 mark

1

- (b) (i) 3 2
for 1 mark

1

- (ii) reversible reaction

so that amount of product depends on conditions used
(linked to first point)

best yield at low temperatures

because it is an exothermic reaction / gives out heat (linked)

reaction rate too slow at low temperatures

450 °C is a compromise between a reasonable yield of ammonia
at a fast rate of reaction

catalyst works best when heated

best yield at high pressures

because there is a decrease in the number of gaseous
molecules (linked)

increasing the pressure also increases the rate

the pressure used is limited by cost, safety etc

the fact that all the nitrogen and hydrogen are not converted to
ammonia does not matter because unreacted gases can be
recycled through process

any six for 1 mark each

6

[10]

7.

- (a) C=C bond in correct position

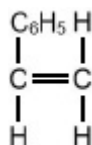
1

3× C-H **and** 1× C-C bond in correct positions

do **not** accept any additional bonds or atoms
 ignore brackets and n before and after displayed
 structural formula

1

an answer of

scores **2** marks

- (b) carboxylic acid (group)

allow carboxyl (group)

1

- (c) water

allow H₂O

1

- (d) (polyester is) thermosoftening

allow (polyester is) thermoplastic

ignore thermoforming

1

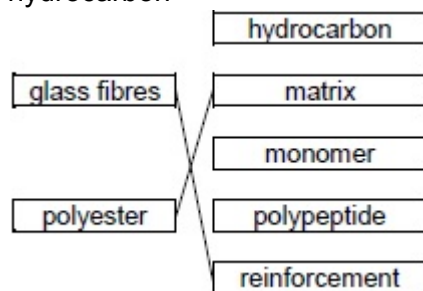
(polyester has) no cross-links

allow intermolecular forces are weak

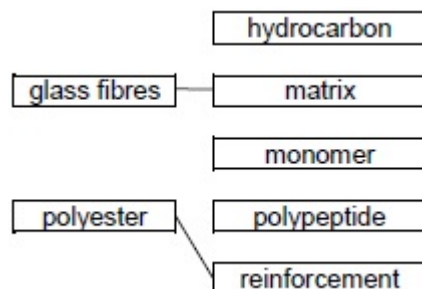
do **not** accept references to breaking covalent bonds or
 breaking chains

1

- (e) hydrocarbon



allow for 1 mark:



2

- (f) any **two** from:
(to make the board)
- harder
 - stronger
 - tougher
 - more rigid
- must be implied comparative statements*
- waterproof

2

[10]

8.

- (a) *allow answers referring
specifically to the naphtha fraction*

crude oil is evaporated/vaporised (by heating)

1

the vapours are condensed (by cooling)

1

(fractions condense) / boil at different temperatures

allow fractions have different boiling points

1

(b) any **four** from:

answer yes or no does not gain credit

ignore references to volume of milk held / number of bottles used / biodegradability / habitats / pollution / mining / dust

each marking point must be a comparison

milk bag points

- uses (75%) less **crude oil** to make (than a plastic milk bottle)
allow eg uses 75% less
poly(ethene) which is made from crude oil
- uses less **energy** / fuel to make (than a plastic / glass milk bottle)
- produces less **carbon dioxide** to manufacture (than a plastic / glass milk bottle)
allow produces less greenhouse gases / causes less global warming
allow produces less CO₂ on burning
- produces less **waste** (than a plastic / glass milk bottle)
allow takes up less landfill (space)
allow an argued case for more waste eg milk bags are discarded / cannot be reused
- less fuel used for **transport** than glass milk bottles
- (produces waste because) milk bags are only used once whereas glass bottles can be **re-used**
allow milk bags are discarded but glass bottles can be reused (24 / many times)
allow glass bottles can be reused but milk bags can't

poly(ethene) points

- uses a limited **raw material** / crude oil whereas the raw materials for glass are almost unlimited
- **less** (5%) poly(ethene) is **recycled** (compared to glass (35%))
allow (35%) glass is recycled or (5%) poly(ethene) (bottles) recycled BUT milk bags aren't / are discarded
or
recycled poly(ethene) is not used to make new bags whereas recycled glass is used to make new bottles

Examiner reports

1.

- (a) Over 90% of students scored both marks, with filtration correctly linked to the removal of solids and sterilisation linked to killing bacteria.
- (b) Nearly all students identified the use of chlorine in sterilisation, but far fewer knew that ozone could also be used.
- (c) Approximately 90% of students identified at least one method of detecting the presence of dissolved solids in water, with 50% identifying both methods.
- (d) The tests for ions are not well known, with very few (roughly 5%) identifying both reagents correctly.
- (e) Most students (nearly 80%) knew that distillation was used to make water pure.

2.

- (a) Sodium hydroxide was well known as the test reagent for identifying aluminium ions. Incorrect reagents, used by themselves or in conjunction with sodium hydroxide, included hydrochloric acid, sodium sulfate, sodium chloride, sodium carbonate, silver nitrate and barium chloride. Other students suggested electrolysis, the use of universal indicator or described boiling the water to leave a residue.

The drinking water / solution does not turn white when sodium hydroxide is added but forms a white precipitate.

Students needed to mention that sodium hydroxide was added in excess somewhere in their response and to be clear that it was the precipitate which dissolves. The precipitate does not go colourless or clear. Terms such as solid / suspension / residue were considered equivalent to precipitate but milkiness / cloudiness were not. Nearly a quarter of students gained all three marks.

- (b) The use of barium chloride in the presence of hydrochloric acid for identifying sulphate ions was well known. Incorrect reagents included using hydrochloric acid alone, barium, silver nitrate or sulfuric acid. $\text{Ba}^{2+}(\text{aq})$ is not a reagent but allowed access to the second mark if stated.

The drinking water / solution does not turn white but forms a white precipitate when barium chloride is added. A white precipitate that redissolves is a contradiction and was not creditworthy. Around 30% of students gained both marks.

- (c) There are several ways to carry out this investigation and there were many creditworthy suggestions. Access to Level 2 was broadly dependent on weighing the empty container (usually a beaker or evaporating basin) and knowing how much water was used (usually "the sample" or a specified volume such as 100 cm^3). Heating to constant mass was often not stated as such but heating to remove all the water was well known.

Some students described distillation and focused on collecting the distillate. Other students described partial evaporation and crystallisation, leaving the dish on a window ledge to allow crystals to form before drying the crystals on a paper towel. About a third of students provided a Level 2 response.

3.

- (a) This was an 'extended response', 'level marked' question. There were three levels and the command word was 'evaluate'. The question was marked holistically. Most answers considered all three relevant stages of a Life Cycle Assessment for both types of cup.

However, many answers failed to consider the environmental impact of the factors discussed. For example, students might say that paper cups are heavier to transport, but did not go on to say what environmental impact that might have in terms of more pollution from lorries, or a greater energy requirement burning more fossil fuels and adding to climate change. This limited their mark to level 2, as did failure to give a strongly linked judgement.

Some students misinterpreted 'recyclable' as 'reusable': the terms are not synonymous. The question stated that both types of cup are disposable, i.e. single use.

Some answers focussed on the economics or ease of use of the cup, rather than environmental impact. Most students were able to acquire at least one mark with 50% of students achieving four marks. 22% of students made it into level 3.

- (b) A mark was often missed through inappropriate rounding to a whole number of cups at an intermediate stage in the working. Others rounded $1000 \div 83$ to 120.5, then multiplied this figure by 550 and quoted an answer to five significant figures. This gave an answer which did not fit the correct value. Had they rounded to two, three or four significant figures they would have achieved the mark. It is never appropriate to quote an answer to more significant figures than an earlier intermediate step. 50% of students achieved both marks.
- (c) Many students tainted an otherwise good answer by referring to intermolecular forces, often alongside a reference to cross-links. They seem not to realise that the cross-links are covalent bonds.

Others were confused and referred to strong bonds or cross-links between monomers. Some referred to layers rather than chains. 21% of students achieved both marks here, although many more gained a mark for 'thermosetting'.

5.

- (a) More than half of the students were able to score three marks, but fewer than 10% gained all five. Most students commented somewhere in the response that both air/oxygen and water are needed for the rusting of iron. The question was generally well answered for test tubes 1, 2 and 3. Some students failed to say whether rusting would be observed even though they had mentioned the conditions within the tube. Test tube 2 was often described as lacking both water and air (because there was a stopper in the tube). Test tube 4 caused some students to introduce the word 'galvanised', others described what would happen if the paint was scratched and some stated that the paint was a sacrificial coating. Few students gained the mark for test tube 5, most saying that the stainless steel nail will not rust because stainless steel does not rust.
- (b) Sacrificial protection was a common answer, however a considerable number of students failed to mention that magnesium is more reactive than iron. Many students appeared to think that the ship was coated in magnesium which behaved as a barrier to corrosive agents or was analogous to galvanising. Nearly 40% gained both marks.
- (c) Many students stated that the lack of corrosion was due to the window frames being painted or covered in plastic. Although the first mark for 'aluminium oxide' was often gained, the second mark, for explaining how this prevents corrosion, did not always follow. Fewer than 20% gained both marks.

7.

- (a) This was generally well answered, with nearly three-quarters of students gaining both marks. Most responses included the double bond in the correct position. A fairly common error was to leave the trailing bonds, resulting in each carbon atom in the double bond having 5 bonds, rather than 4. The brackets and 'n' from the polymer structure were often retained, and although not penalised, this practice should be discouraged.
- (b) This was also generally well answered, with nearly two-thirds of students gaining this mark. Where students did not know the name of the carboxylic acid group their incorrect response was often 'alcohol'.
- (c) Most responses correctly named the small molecule released in condensation polymerisation as water, although a significant minority opted for ester. Over two-thirds of students gained this mark.
- (d) About 50% of pupils either did not know, or could not recall the term 'thermosoftening'. A significant minority referred to 'thermoforming'. For the second mark, responses often talked about weak intermolecular forces but then went on to incorrectly state these were 'between monomers'. Other responses vaguely talked about 'weak forces' without any further clarification. References to weak covalent bonds were also often seen. Nearly a quarter of students gained both marks.
- (e) A significant minority of students identified the two components of a composite material but confused them in this context. Such students were awarded one mark. Around a quarter of students gained both marks.
- (f) Describing the consequences of the properties of the surfboard, rather than focusing on the properties themselves were the most common responses which were not credited in this question. Only just over 10% were able to give two creditworthy properties.

8.

- (a) This question was a good discriminator. The explanation of how fractions, such as naphtha, are separated from crude oil was generally well understood. The better students gained three marks for describing that crude oil is heated; this causes evaporation, and as the vapours rise up the fractionating column they condense at a number of different temperatures.
- (b) Good students gave clear advantages and clear disadvantages, and made appropriate comparisons of the environmental impact of using milk bags instead of milk bottles. Weaker students were unable to do more than simply repeat information already given in the article. These students also often used basic statements about possible environmental issues, such as, 'recyclable', 'cause pollution' and 'produce carbon dioxide' without a link to the type of milk container. The biggest problem was that many students just listed the advantages and disadvantages without mentioning the comparisons that the question required.